

## 35.1 Logistic regression

A binary response records whether something happened: a patient recovered, a voter turned out, a loan defaulted. Then  $E(Y_i) = \Pr(Y_i = 1)$ , so modelling the mean *is* modelling a probability, and a linear model for it is in trouble before it starts:  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  ranges over the whole line while a probability may not leave  $[0, 1]$ , and the variance  $\pi_i(1 - \pi_i)$  moves with the mean. The generalized linear model of [Chapter 34](#) removes both problems by putting a link between the mean and the linear predictor.

### 35.1.1 The model

Let  $Y_1, \dots, Y_n$  be independent with  $Y_i \sim \text{Binomial}(m_i, \pi_i)$ ;  $m_i = 1$  throughout is **ungrouped** data, and larger  $m_i$  arise when several units share a covariate vector. Dropping the index,

$$\binom{m}{y} \pi^y (1 - \pi)^{m-y} = \binom{m}{y} \exp \left\{ y \log \frac{\pi}{1 - \pi} + m \log(1 - \pi) \right\},$$

an exponential dispersion family (Definition 34.1.1) for the mean  $\mu = m\pi$ , with natural parameter  $\theta = \log\{\pi/(1 - \pi)\}$ , cumulant function  $b(\theta) = \log(1 + e^\theta)$ , dispersion  $\phi = 1$  and weight  $m$ . Differentiating  $b$  twice recovers the moments of Proposition 34.1.2:  $b'(\theta) = \pi$  and  $b''(\theta) = \pi(1 - \pi)$ , so the variance function is  $V(\pi) = \pi(1 - \pi)$  and  $\text{Var}(Y) = m\pi(1 - \pi)$ . The natural parameter is the **log odds**, and the canonical link is therefore the logit.

#### Definition 35.1.1 — Logistic regression

The **logistic regression model** for independent  $Y_i \sim \text{Binomial}(m_i, \pi_i)$  is

$$\text{logit } \pi_i = \log \frac{\pi_i}{1 - \pi_i} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \text{equivalently} \quad \pi_i = \frac{\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}. \quad (35.1.1)$$

The quantity  $\pi/(1 - \pi)$  is the **odds** of the event, and the ratio of the odds at two covariate vectors is an **odds ratio**.

The logit is one choice among many ([Section 35.2](#)). Its claim to first place is that it is canonical, so Theorem 34.3.1 takes its simplest form, and that its coefficients say something people want to hear.

### 35.1.2 Reading the coefficients

#### Proposition 35.1.1 — Three readings of a logistic coefficient

In [Eq. 35.1.1](#), write  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  and let  $x_j$  be a covariate that appears in the linear predictor only through its own coefficient  $\beta_j$ . Then:

- (a) increasing  $x_j$  by  $c$  with the other covariates unchanged multiplies the odds by  $e^{c\beta_j}$ ,

whatever the values of the other covariates;

- (b)  $\partial\pi/\partial x_j = \beta_j \pi(1 - \pi)$ , so  $|\partial\pi/\partial x_j| \leq |\beta_j|/4$ , with equality exactly at  $\pi = 1/2$ ;
- (c) the average of that derivative over the sample,  $n^{-1} \sum_i \beta_j \hat{\pi}_i(1 - \hat{\pi}_i)$ , is the **average marginal effect**: the mean change in fitted probability per unit of  $x_j$ .

*Proof.*

- (a) The log odds at  $x_j + c$  minus the log odds at  $x_j$  is  $c\beta_j$ , and everything else in  $\eta$  cancels. (b) Differentiating  $\pi = e^\eta/(1 + e^\eta)$  gives  $d\pi/d\eta = \pi(1 - \pi)$ , and the chain rule supplies the factor  $\beta_j$ . The function  $\pi(1 - \pi)$  on  $[0, 1]$  is maximized at  $\pi = 1/2$  with value  $1/4$ , so the derivative has modulus at most  $|\beta_j|/4$ . (c) Immediate from (b).

■

Part (a) is why logistic regression is written as it is: one number summarizes the effect on the odds at every setting of the other covariates. Part (b) is Gelman and Hill's "divide by four" rule, sharp near  $\pi = 1/2$ ; part (c) reports the same coefficient on the probability scale.

*Warning.* None of the three readings is causal: each compares two covariate vectors *within the fitted model*, and whether that estimates an intervention is the question of [Chapter 25](#). [Section 35.5](#) adds a wrinkle special to odds ratios.

### Example. (Voting intention in the 1996 American election)

The 1996 American National Election Studies survey recorded, for 944 respondents, the candidate they expected to vote for (393 said Dole), party identification from strong Democrat (0) to strong Republican (6), age, education on a seven-point scale and household income on a 24-point scale. Fitting [Eq. 35.1.1](#) with these four covariates and age in decades gives

| term                 | $\hat{\beta}_j$ | standard error |
|----------------------|-----------------|----------------|
| intercept            | -5.471          | 0.619          |
| party identification | 1.2188          | 0.0709         |
| age (decades)        | 0.1222          | 0.0714         |
| education            | 0.0147          | 0.0771         |
| income               | 0.0311          | 0.0210         |

One step along the party scale multiplies the odds of a Dole vote by  $e^{1.2188} = 3.383$ ; three steps multiply them by 38.72. Dividing by four bounds the effect on the probability by 0.305 per step, and the average marginal effect is 0.100: one step is worth about ten percentage points, averaged over the respondents ([Figure 35.1.1](#)).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
```

```
anes = sm.datasets.anes96.load_pandas().data
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
```

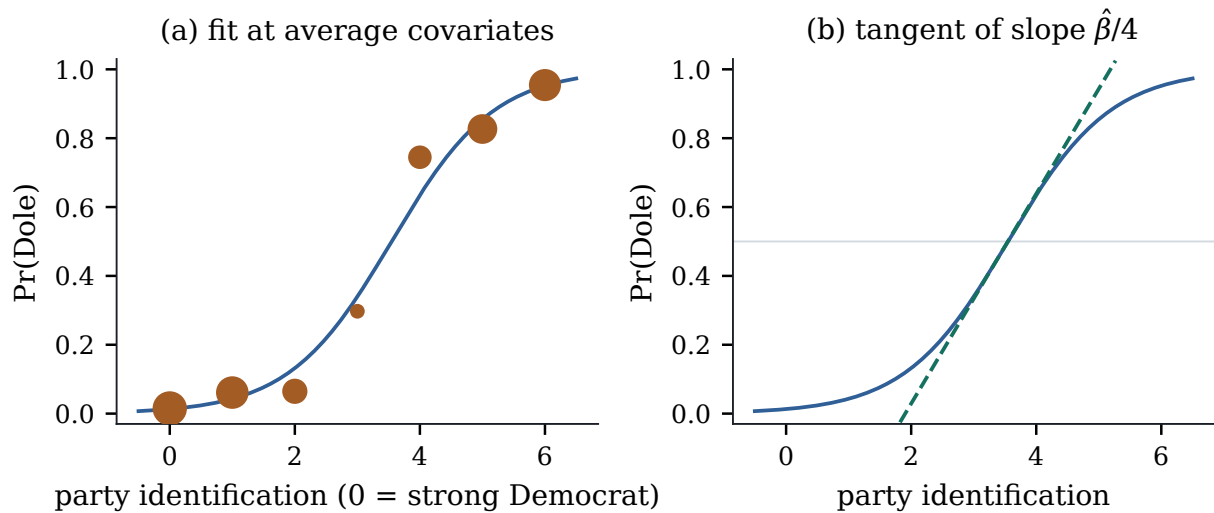


Figure 35.1.1: Expected vote for Dole in the 1996 ANES survey. (a) The fitted logistic curve against party identification, the other covariates held at their means; the discs are the observed proportions in the seven party groups, with area proportional to group size. (b) The tangent at the steepest point, of slope  $\hat{\beta}_j/4$ .

```

                                "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"]                                     # 1 = Dole, 0 = Clinton

fit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
pi_hat = fit.fittedvalues

print(fit.params.round(4).to_string())
print("odds ratio for one step on the party scale:", round(np.exp(fit.params["PID"])
, 3))
print("divide by four:", round(fit.params["PID"] / 4, 3))
print("average marginal effect:", round((fit.params["PID"] * pi_hat * (1 - pi_hat)).
mean(), 3))

```

[Run this code online](#)

### 35.1.3 Grouped and ungrouped data

Two data sets recording the same information can look very different: one lists 944 individual zeros and ones, the other one binomial count per distinct covariate vector.

#### Proposition 35.1.2 — Grouping

Let  $\mathbf{y}$  be ungrouped binary data, and suppose the rows of  $\mathbf{X}$  take  $N$  distinct values  $\mathbf{x}_{(1)}^*, \dots, \mathbf{x}_{(N)}^*$ , the  $k$ th occurring  $m_k$  times with  $s_k$  successes. Then:

- the two log-likelihoods differ by a constant free of  $\boldsymbol{\beta}$ , so they have the same maximizers, the same score, the same information and the same likelihood ratio statistics;
- the two deviances of Definition 34.5.1 differ by  $-2\ell_{\text{sat}}$ , where  $\ell_{\text{sat}} = \sum_k \{s_k \log(s_k/m_k) + (m_k - s_k) \log(1 - s_k/m_k)\}$  is the maximized log-likelihood of the grouped saturated

model;

- (c) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\sum_i m_i \hat{\pi}_i = \sum_i y_i$  in either form: the fit reproduces the total number of successes.

*Proof.*

- (a) With  $\eta_k = \mathbf{x}_{(k)}^* \boldsymbol{\beta}$ , the ungrouped log-likelihood is

$$\sum_{k=1}^N \{s_k \eta_k - m_k \log(1 + e^{\eta_k})\},$$

because the individual contributions within a group depend on the data only through  $s_k$ . The grouped log-likelihood is the same expression plus  $\sum_k \log \binom{m_k}{s_k}$ , a constant. (b) The deviance is twice the difference between the saturated and fitted log-likelihoods. The fitted log-likelihood is the same in both forms by (a); the ungrouped saturated model sets  $\hat{\pi}_i = y_i$  and attains log-likelihood 0, while the grouped saturated model sets  $\hat{\pi}_k = s_k/m_k$  and attains  $\ell_{\text{sat}} \leq 0$ . (c) Under the canonical link the likelihood equations are  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$  (Theorem 34.3.1); the column  $\mathbf{1}$  gives  $\sum_i (y_i - m_i \hat{\pi}_i) = 0$ .

■

Part (b) is worth pausing over: the deviance of a logistic fit depends on how the data were typed in. *Differences* of deviances between nested models do not, because the saturated term cancels, and those are what Theorem 34.5.5 licenses.

### Example. (The same fit, two deviances)

Fit logit  $\pi = \beta_0 + \beta_1(\text{party}) + \beta_2(\text{education})$  to the survey of [Example \(Voting intention in the 1996 American election\)](#). The 944 respondents occupy 47 distinct covariate cells. Grouped, the deviance is 45.99 on 44 degrees of freedom; ungrouped, it is 533.34 on 941. The coefficients and their standard errors agree to machine precision.

```
# the same model in grouped form: one binomial count per distinct covariate pattern
cols = ["PID", "educ"]
cells = anes.groupby(cols)["vote"].agg(["sum", "count"]).reset_index()
Xg = sm.add_constant(cells[cols])
counts = np.column_stack([cells["sum"], cells["count"] - cells["sum"]])
grouped = sm.GLM(counts, Xg, family=sm.families.Binomial()).fit()

Xu = sm.add_constant(anes[cols])
ungrouped = sm.GLM(anes["vote"], Xu, family=sm.families.Binomial()).fit()

print("grouped  ", grouped.params.round(4).to_numpy(), "deviance", round(grouped.
    deviance, 2))
print("ungrouped", ungrouped.params.round(4).to_numpy(), "deviance", round(ungrouped
    .deviance, 2))
```

[Run this code online](#)

### 35.1.4 Estimation

There is no closed form for  $\hat{\boldsymbol{\beta}}$ . Under the canonical link Theorem 34.3.1 reduces to

$$\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}, \quad \mu_i = m_i \pi_i = \frac{m_i \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}, \quad (35.1.2)$$

a system of  $p$  smooth equations in  $p$  unknowns. Specializing Theorem 34.4.1 gives the algorithm every implementation uses.

#### Proposition 35.1.3 — Fisher scoring for logistic regression

For Eq. 35.1.1 the Fisher scoring step from  $\mathbf{b}$  is the weighted least squares fit of the working response  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\mathbf{W}$ , where

$$w_i = m_i \pi_i (1 - \pi_i), \quad z_i = \eta_i + \frac{y_i - m_i \pi_i}{m_i \pi_i (1 - \pi_i)},$$

both evaluated at  $\mathbf{b}$ . The observed and expected information coincide and equal  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ ; the log-likelihood is concave in  $\boldsymbol{\beta}$ , and strictly concave when  $\text{rank}(\mathbf{X}) = p$  and  $0 < \pi_i < 1$ .

*Proof.* The weights and working response are the general formulae of Theorem 34.4.1 with  $V(\pi) = \pi(1 - \pi)$  and  $d\eta/d\mu = 1/\{m\pi(1 - \pi)\}$ . Differentiating Eq. 35.1.2 once more gives the Hessian  $-\mathbf{X}^\top \mathbf{W} \mathbf{X}$ , so the observed information equals the expected information, as Theorem 34.3.1 gives for every canonical link. The matrix  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  is nonnegative definite, and positive definite when the  $w_i$  are positive and  $\mathbf{X}$  has full column rank, since  $\mathbf{b}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{b} = \sum_i w_i (\mathbf{x}_{(i)}^\top \mathbf{b})^2$  vanishes only if  $\mathbf{X} \mathbf{b} = \mathbf{0}$ . ■

Concavity means at most one local maximum, and the step is the weighted least squares of Theorem 21.3.1 with response and weights refreshed each iteration. Convergence from  $\boldsymbol{\beta} = \mathbf{0}$  is usually fast, and took 6 iterations in [Example \(Voting intention in the 1996 American election\)](#); it is not guaranteed, and [Section 35.3](#) describes the configuration in which there is nothing to converge to.

### 35.1.5 Inference

With the estimate in hand, Theorem 34.5.1 supplies its law: under the usual regularity conditions, with  $p$  fixed and the information growing without bound,  $\hat{\boldsymbol{\beta}}$  is treated as  $N_p(\boldsymbol{\beta}, (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1})$ . The growth condition is real: it fails if a covariate is almost constant, and it fails if the fitted probabilities are driven to 0 or 1, since then  $w_i = m_i \pi_i (1 - \pi_i) \rightarrow 0$ . Of the three statistics of Proposition 34.5.3, the Wald statistic is what software prints, and it is the one the Hauck–Donner effect spoils; the likelihood ratio statistic is invariant and survives an infinite estimate, and the score statistic needs only the null fit.

**Example. (Tests and intervals in the voting model)**

In [Example \(Voting intention in the 1996 American election\)](#), education has  $\hat{\beta} = 0.0147$  with standard error 0.0771; its Wald statistic is 0.036 and so is the likelihood ratio statistic from dropping it, while age has  $z = 1.71$  and  $p = 0.087$ . For the party coefficient the Wald interval is (1.080, 1.358) and the profile likelihood interval (1.086, 1.364): the log-likelihood is nearly quadratic here. [Example \(A separated subgroup\)](#) shows what happens when it is not.

The machinery of Part III survives in weakened form: simultaneous inference uses a quadratic form in  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  with matrix  $\{\mathbf{\Lambda}^\top (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{\Lambda}\}^{-1}$ , referred to  $\chi^2(q)$  in place of the  $F$  statistic of Theorem 11.3.2. What does *not* survive is exactness. The right measure of how much there is to work with is the total weight  $\sum_i m_i \pi_i (1 - \pi_i)$ : a study of 500 subjects with twelve events has total weight about 12, the same as a balanced study of fifty, which is the arithmetic behind the events-per-variable rules of thumb in the applied literature.

**35.1.6 Exercises****35.1.6.1 A. Check your understanding****Exercise A1.**

A logistic model has  $\hat{\beta} = 0.7$  for a binary treatment indicator. Give the estimated odds ratio, the divide-by-four bound on the change in probability, and the actual change when the control probability is 0.5, 0.1 and 0.01.

*Solution.* The odds ratio is  $e^{0.7} = 2.014$  and the bound is  $0.7/4 = 0.175$ . Starting from odds 1, 1/9 and 1/99, the treated odds are 2.014, 0.2238 and 0.02034, so the probabilities become 0.668, 0.183 and 0.0199 and the changes are 0.168, 0.083 and 0.0099. The bound is nearly attained at  $\pi = 1/2$  and grossly overstates the change for rare events.

**Exercise A2.**

Show from [Eq. 35.1.2](#) that a logistic model containing an indicator column for a group reproduces that group's observed proportion exactly. What follows for a model whose covariates are a full set of indicators for a categorical variable?

**35.1.6.2 B. Practice****Exercise B1.**

In logit  $\pi = \beta_0 + \beta_1 x$  with  $\beta_1 \neq 0$ , the level at which  $\pi = 1/2$  is  $x_{1/2} = -\beta_0/\beta_1$  (in bioassay, the median effective dose). Derive the delta-method standard error of  $\hat{x}_{1/2}$ , and explain why it becomes useless as  $\hat{\beta}_1$  approaches zero.

**Exercise B2.**

For grouped data with all  $m_i$  large, Exercise (B2) fits  $\boldsymbol{\beta}$  by weighted least squares on the empirical logits  $\ell_i = \text{logit}(s_i/m_i)$  with weights  $m_i\hat{\pi}_i(1 - \hat{\pi}_i)$ . Show that one Fisher scoring step of [Proposition 35.1.3](#), started from the fit that reproduces the observed proportions, is exactly that estimate.

*Solution.* Start from a  $\boldsymbol{\beta}$  with  $\pi_i = s_i/m_i$ , so  $\eta_i = \ell_i$ . The working residual  $(y_i - m_i\pi_i)/\{m_i\pi_i(1 - \pi_i)\}$  vanishes, the working response is  $z_i = \ell_i$  and the weights are  $m_i\hat{\pi}_i(1 - \hat{\pi}_i)$ , so one step returns the weighted least squares fit. The empirical-logit fit is therefore a legitimate starting value; it is undefined when  $s_i \in \{0, m_i\}$ .

**Exercise B3.**

Two binary covariates enter with an interaction:  $\text{logit } \pi = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ . Show that  $\beta_{12}$  is the log of a ratio of two odds ratios, and that  $\beta_{12} = 0$  does *not* make the effect of  $x_1$  on the probability scale the same at both levels of  $x_2$ .

**35.1.6.3 C. Going deeper****Exercise C1.**

Prove that  $\ell(\boldsymbol{\beta}) = -\sum_i \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$ , with  $s_i = 2y_i - 1$ , is concave in  $\boldsymbol{\beta}$ , without computing the Hessian.

*Solution.* The map  $u \mapsto \log(1 + e^u)$  is convex, since its second derivative  $e^u/(1 + e^u)^2$  is positive. A convex function composed with an affine map is convex, so each  $\boldsymbol{\beta} \mapsto \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$  is convex, and a sum of convex functions is convex. Hence  $\ell$  is concave.